

BIOLOGICAL BATTLE ROYALE: GENE REGULATORY NETWORKS THAT SUPPORT MULTI-FATE CELLULAR DECISIONS

MS: BOOLEAN MODELS IN SYSTEMS BIOLOGY, ECMTB 2026, GRAZ

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INTRODUCTION

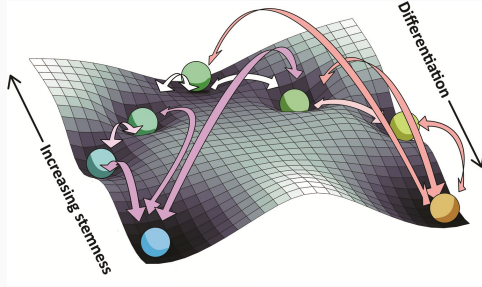
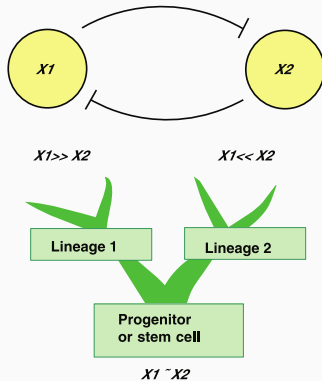


Figure: Waddington's Landscape¹

- Cell fate determination $\equiv$ Waddington's landscape
- Gene regulatory networks (GRNs) regulate trajectories
- Master regulators - differentiation to particular state
- Fully differentiated - Exclusive expression (Single high)

¹Furusawa and Kaneko, *Science*, 2012; Sáez et al., *Interface Focus*, 2022. Figure by Atchuta Srinivas Duddu

BINARY CELL FATE DECISION



Cells	Transcription factors		Cell fates		
	X1	X2	$X1 > X2$	$X1 \sim X2$	$X1 < X2$
Early embryo	Cdx2	Oct4	Trophectoderm	Totipotent embryo	Inner cell mass
Embryo ICM	GATA6	Nanog	Primitive endoderm	Inner cell mass	Epiblast
Blood	GATA1	PU.1	Erythroid cells	Common myeloid progenitor	Myeloid cells
Pancreas	Ptf1a	Nkx6	Exocrine cells	Pancreatic progenitor	Endocrine cells
Somite	Pax3	Foxc2	Myogenic cells	Dermomyotome progenitor	Vascular cells

Figure: Toggle switch captures binary branching of cell fate¹

¹Zhou and Huang, *Trends in genetics: TIG.*, 2011

EXTENSION TO N-TERMINAL STATES

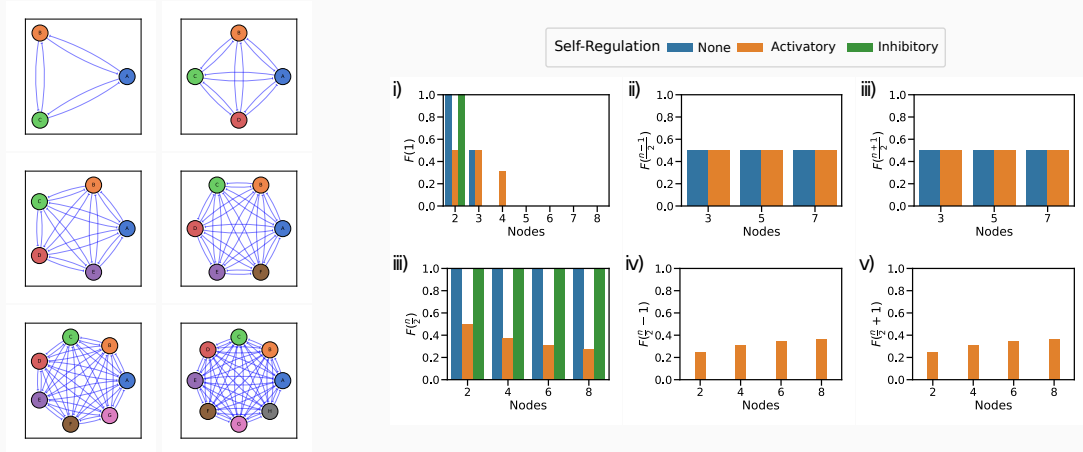


Figure: Toggle-n networks¹

Figure: Toggle-n gives $n/2$ -high steady states¹

¹Bv et al., *Journal of The Royal Society Interface*, 2025

WHICH NETWORKS CAN GIVE RISE TO FULLY DIFFERENTIATED STATES?

SINGLE HIGH STATES - ONLY ONE MASTER REGULATOR IS HIGH

METHODS

MONOTONE BOOLEAN FUNCTIONS (MBFs)

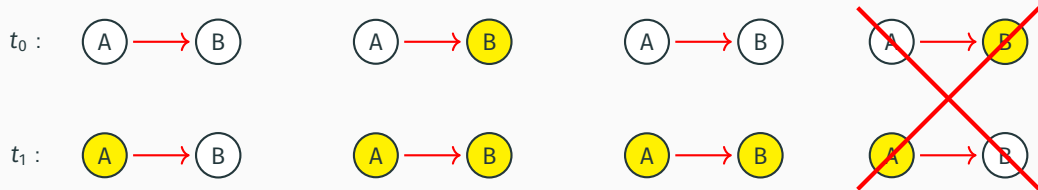


Figure: Schematic of increasing MBF(1) for activatory input

¹Korshunov, *Russian Mathematical Surveys*, 2003

MONOTONE BOOLEAN FUNCTIONS (MBFs)

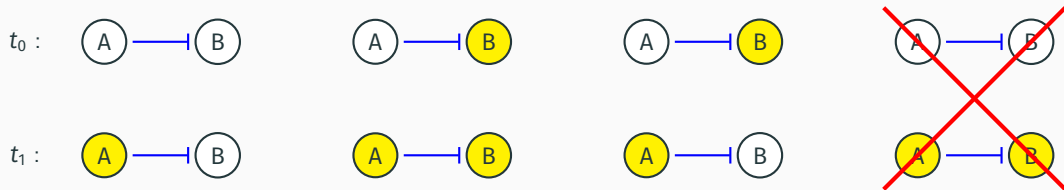
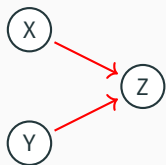


Figure: Schematic of decreasing MBF(1) for inhibitory input

¹Korshunov, *Russian Mathematical Surveys*, 2003



Inputs		MBFs					
X	Y	0	$\wedge$	X	Y	$\vee$	1
0	0	0	0	0	0	0	1
0	1	0	0	0	1	1	1
1	0	0	0	1	0	1	1
1	1	0	1	1	1	1	1

Figure: Table of MBF(2) for a node with two activatory inputs X and Y.

¹Korshunov, *Russian Mathematical Surveys*, 2003

$$U(b) := \{f \in MBF(s) \mid f(b) = 1\}$$

$$L(b) := \{f \in MBF(s) \mid f(b) = 0\}.$$

¹ $b \in \mathbb{B}^s$; s = No. of inputs; Dashed lines indicate prime ideals, Red lines indicate prime filters

²Adigwe et al., *PLOS Complex Systems*, 2026

$$U(b) := \{f \in MBF(s) \mid f(b) = 1\}$$

$$L(b) := \{f \in MBF(s) \mid f(b) = 0\}.$$

L(11)	$\{\mathbf{0}\}$	1
L(10)	$\{\mathbf{0}, \wedge, Y\}$	3
L(01)	$\{\mathbf{0}, \wedge, X\}$	3
L(00)	$\{\mathbf{0}, \wedge, X, Y, \vee\}$	5
U(11)	$\{\mathbf{1}, \vee, X, Y, \wedge\}$	5
U(10)	$\{\mathbf{1}, \vee, X\}$	3
U(01)	$\{\mathbf{1}, \vee, Y\}$	3
U(00)	$\{\mathbf{1}\}$	1

Figure: U and L for $MBF(2)$

¹ $b \in \mathbb{B}^s$; s = No. of inputs; Dashed lines indicate prime ideals, Red lines indicate prime filters

²Adigwe et al., *PLOS Complex Systems*, 2026

$$U(b) := \{f \in MBF(s) \mid f(b) = 1\}$$

$$L(b) := \{f \in MBF(s) \mid f(b) = 0\}.$$

Lattice theory:

- U s = prime ideals
- L s = prime filters

L(11)	$\{0\}$	1
L(10)	$\{0, \wedge, Y\}$	3
L(01)	$\{0, \wedge, X\}$	3
L(00)	$\{0, \wedge, X, Y, \vee\}$	5
U(11)	$\{1, \vee, X, Y, \wedge\}$	5
U(10)	$\{1, \vee, X\}$	3
U(01)	$\{1, \vee, Y\}$	3
U(00)	$\{1\}$	1

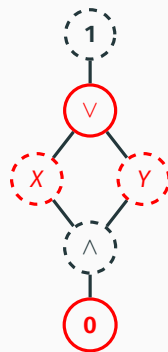


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²Adigwe et al., *PLOS Complex Systems*, 2026

Network $N = (V, E, \delta)$

Boolean models $f = (f_1, f_2, \dots, f_n)$

Boolean state $b = (b_1, \dots, b_n) \in \mathbb{B}^n$

Steady state $\implies f(b) = b$

$$f_i \in I_i(\beta_i(b))$$

where,

$$I_i(b) = \begin{cases} U(\beta_i(b_{\mathbf{S}(i)})) & \text{if } b_i = 1 \\ L(\beta_i(b_{\mathbf{S}(i)})) & \text{if } b_i = 0 \end{cases}$$

$$\beta_i^j = \begin{cases} Id & \text{if } \delta(j, i) = 1 \\ \neg Id & \text{if } \delta(j, i) = -1 \end{cases}.$$

¹ $b_{\mathbf{S}(i)}$ = values of the inputs of node i evaluated at the state b

²Adigwe et al., *PLOS Complex Systems*, 2026

Steady states: $b^1, \dots, b^q \in \mathbb{B}^n$

$$f_i \in \bigcap_{j=1}^q I_i(b^j)$$

where,

$$I_i(b^k) = \begin{cases} U(\beta_i(b_{\mathbf{s}(i)}^k)) & \text{if } b_i^k = 1 \\ L(\beta_i(b_{\mathbf{s}(i)}^k)) & \text{if } b_i^k = 0 \end{cases}$$

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²Adigwe et al., *PLOS Complex Systems*, 2026

RESULTS

WHICH NETWORK(S) HAVE THE MAXIMUM PREVALENCE OF SINGLE HIGH STATES?

Most MBMs for $e \in \mathbb{B}^n$

$$H = \prod_{i=1}^n (|\text{MBF}(s_i)| - 1).$$

$$|L(\mathbf{0}^S)| = |U(\mathbf{1}^S)| = |\text{MBF}(s)| - 1$$

¹ s_i = No. of inputs to node i ; $\mathbf{0}^S(\mathbf{1}^S)$ = vectors of zeros(ones) $\in \mathbb{B}^S$; e^k = k -th unit vector $\in \mathbb{B}^n$

²Adigwe et al., *PLOS Complex Systems*, 2026

³Bv et al., *bioRxiv*, 2026

TEAM NETWORKS MAXIMIZE PREVALANCE OF ONE SINGLE HIGH STATE

Most MBMs for $e \in \mathbb{B}^n$

$$H = \prod_{i=1}^n (|\text{MBF}(s_i)| - 1).$$

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$e^1 = (10 \dots 0)$ with H MBMs

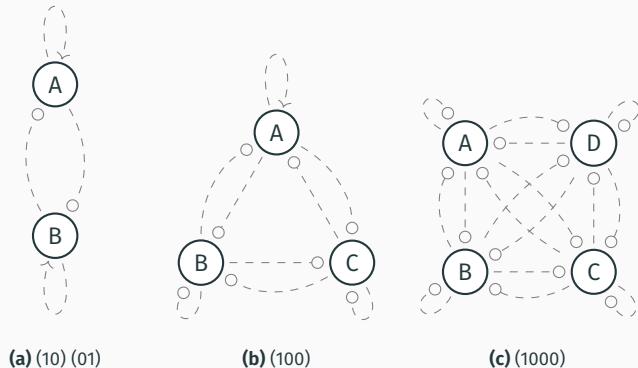


Figure: Network structures with H MBMs supporting e^1

¹ s_i = No. of inputs to node i ; $\mathbf{0}^S(\mathbf{1}^S)$ = vectors of zeros(ones) $\in \mathbb{B}^S$; e^k = k -th unit vector $\in \mathbb{B}^n$

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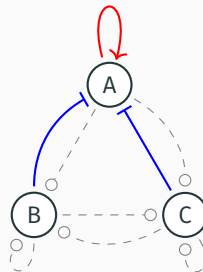
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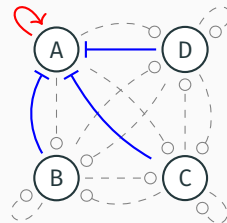
$$U(\beta_1(10 \dots 0)) \rightarrow U(11 \dots 1)$$



(a) (10)



(b) (100)



(c) (1000)

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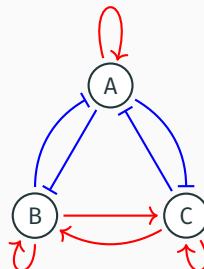
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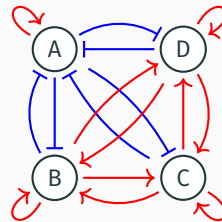
$$L(\beta_j(10 \dots 0)) \rightarrow L(00 \dots 0)$$



(a) (10)



(b) (100)



(c) (1000)

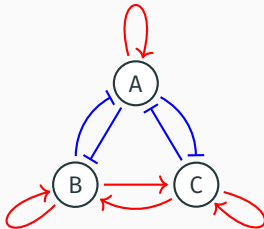
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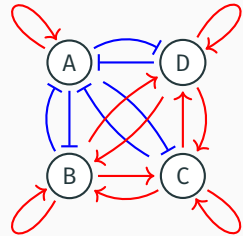
²Adigwe et al., *PLOS Complex Systems*, 2026

³Bv et al., *bioRxiv*, 2026

e^i with H MBMs



(a) (100)(010)(001)



(b) (1000)(0100)(0010)(0001)

Figure: Network structures with H MBMs supporting $e^1, \dots, e^n$

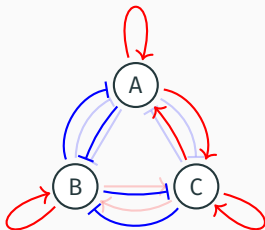
¹ $e^k = k$ -th unit vector $\in \mathbb{B}^n$

²Bv et al., *bioRxiv*, 2026

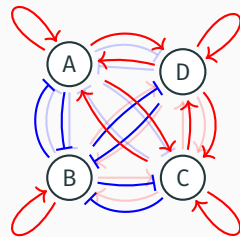
DISCONNECTED NETWORKS MAXIMIZE ALL SINGLE HIGH STATES

e^i with H MBMs

e^j with H MBMs



(a) $(100)(010)(001)$



(b) $(1000)(0100)(0010)(0001)$

Figure: Network structures with H MBMs supporting $e^1, \dots, e^n$

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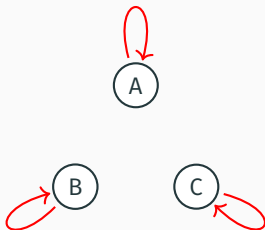
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e^i with H MBMs

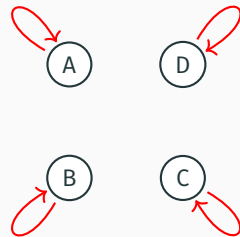
e^j with H MBMs

Disconnected networks

Domain of attraction(e^i) = $\{e^i\}$



(a) (100)(010)(001)



(b) (1000)(0100)(0010)(0001)

Figure: Network structures with H MBMs supporting $e^1, \dots, e^n$

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²Bv et al., *bioRxiv*, 2026

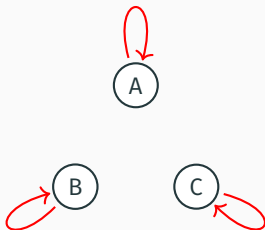
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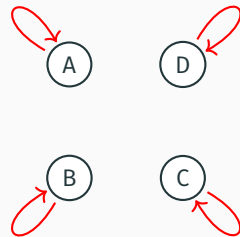
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Figure: Network structures with H MBMs supporting $e^1, \dots, e^n$

Network without mutual communication cannot serve as decision network

¹ e^k = k -th unit vector $\in \mathbb{B}^n$

²Bv et al., *bioRxiv*, 2026

**WHICH NETWORK(S) HAVE EQUAL PREVALANCE OF SINGLE HIGH STATES?
(EQUIPOTENT NETWORKS)**

WHICH AMONG THEM HAVE THE MOST NUMBER OF MODELS?

	f_1	f_2	$\dots$	f_n
e^1	$ U(\beta_1(e_{\mathbf{s}(1)}^1)) $	$ L(\beta_2(e_{\mathbf{s}(2)}^1)) $	$\dots$	$ L(\beta_n(e_{\mathbf{s}(n)}^1)) $
e^2	$ L(\beta_1(e_{\mathbf{s}(1)}^2)) $	$ U(\beta_2(e_{\mathbf{s}(2)}^2)) $	$\dots$	$ L(\beta_n(e_{\mathbf{s}(n)}^2)) $
$\vdots$	$\vdots$	$\vdots$	$\ddots$	$\vdots$
e^n	$ L(\beta_1(e_{\mathbf{s}(1)}^n)) $	$ L(\beta_2(e_{\mathbf{s}(2)}^n)) $	$\dots$	$ U(\beta_n(e_{\mathbf{s}(n)}^n)) $

¹ $b_{\mathbf{s}(i)}$ = values of the inputs of node i evaluated at the state b ; $\mathbf{0}^S(\mathbf{1}^S)$ = vectors of zeros(ones) $\in \mathbb{B}^S$; $e^{j,S}$ = j -th unit vector $\in \mathbb{B}^S$

²Bv et al., *bioRxiv*, 2026

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e^2	$ L(\beta_1(e_{\mathbf{s}(1)}^2)) $	$ U(\beta_2(e_{\mathbf{s}(2)}^2)) $	$\dots$	$ L(\beta_n(e_{\mathbf{s}(n)}^2)) $
$\vdots$	$\vdots$	$\vdots$	$\ddots$	$\vdots$
e^n	$ L(\beta_1(e_{\mathbf{s}(1)}^n)) $	$ L(\beta_2(e_{\mathbf{s}(2)}^n)) $	$\dots$	$ U(\beta_n(e_{\mathbf{s}(n)}^n)) $

All row-wise products should be equal

¹ $b_{\mathbf{s}(i)}$ = values of the inputs of node i evaluated at the state b ; $\mathbf{0}^S(\mathbf{1}^S)$ = vectors of zeros(ones) $\in \mathbb{B}^S$; $e^{j,S}$ = j -th unit vector $\in \mathbb{B}^S$

²Bv et al., *bioRxiv*, 2026

Consider networks with all-to-all coupling without self-edges

	f_1	f_2	$\dots$	f_n
e^1	$ U(\beta_1(\mathbf{0}^{n-1})) $	$ L(\beta_2(e^{1,n-1})) $	$\dots$	$ L(\beta_n(e^{1,n-1})) $
e^2	$ L(\beta_1(e^{1,n-1})) $	$ U(\beta_2(\mathbf{0}^{n-1})) $	$\dots$	$ L(\beta_n(e^{2,n-1})) $
$\vdots$	$\vdots$	$\vdots$	$\ddots$	$\vdots$
e^n	$ L(\beta_1(e^{n-1,n-1})) $	$ L(\beta_2(e^{n-1,n-1})) $	$\dots$	$ U(\beta_n(\mathbf{0}^{n-1})) $

All row-wise products should be equal

¹ $b_{S(i)}$ = values of the inputs of node i evaluated at the state b ; $\mathbf{0}^S(\mathbf{1}^S)$ = vectors of zeros(ones) $\in \mathbb{B}^S$; $e^{j,S}$ = j -th unit vector $\in \mathbb{B}^S$

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	f_1	f_2	$\dots$	f_n
e^1	$ U(\beta_1(\mathbf{0}^{n-1})) $	$ L(\beta_2(e^{1,n-1})) $	$\dots$	$ L(\beta_n(e^{1,n-1})) $
e^2	$ L(\beta_1(e^{1,n-1})) $	$ U(\beta_2(\mathbf{0}^{n-1})) $	$\dots$	$ L(\beta_n(e^{2,n-1})) $
$\vdots$	$\vdots$	$\vdots$	$\ddots$	$\vdots$
e^n	$ L(\beta_1(e^{n-1,n-1})) $	$ L(\beta_2(e^{n-1,n-1})) $	$\dots$	$ U(\beta_n(\mathbf{0}^{n-1})) $

All row-wise products should be equal

Column values for node i with m inhibitions

$$q(m) := \left\{ \kappa_m, \underbrace{\kappa_{n-m}, \dots, \kappa_{n-m}}_{m \text{ times}}, \underbrace{\kappa_{n-m-2}, \dots, \kappa_{n-m-2}}_{n-m-1 \text{ times}} \right\}.$$

$$\kappa_k \in K_S$$

$$K_2 = \{1, 3, 5\}$$

$$K_3 = \{1, 6, 14, 19\}$$

$$K_4 = \{1, 20, 84, 148, 167\}$$

¹ $b_{S(i)}$ = values of the inputs of node i evaluated at the state b ; $\mathbf{0}^S(\mathbf{1}^S)$ = vectors of zeros(ones) $\in \mathbb{B}^S$; $e^{j,S}$ = j -th unit vector $\in \mathbb{B}^S$

²Bv et al., *bioRxiv*, 2026

EQUIPOTENT 3-NODE NETWORK ARE SYMMETRIC

m	$q(m)$		$\prod$
0	$\{\kappa_0, \kappa_1, \kappa_1\}$	$\{1, 3, 3\}$	9
1	$\{\kappa_1, \kappa_2, \kappa_0\}$	$\{3, 5, 1\}$	15
2	$\{\kappa_2, \kappa_1, \kappa_1\}$	$\{5, 3, 3\}$	45

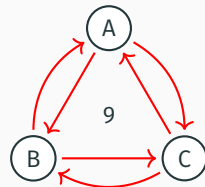
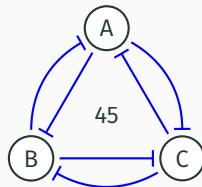


Figure: 3-node equipotent networks (No. of MBMs supporting each single high state indicated)

¹ m = No. of inhibitions; $q(m)$ = Columns values; $\prod$ = row-wise product

²Bv et al., *bioRxiv*, 2026

EQUIPOTENT 3-NODE NETWORK ARE SYMMETRIC

m	$q(m)$		$\prod$
0	$\{\kappa_0, \kappa_1, \kappa_1\}$	$\{1, 3, 3\}$	9
1	$\{\kappa_1, \kappa_2, \kappa_0\}$	$\{3, 5, 1\}$	15
2	$\{\kappa_2, \kappa_1, \kappa_1\}$	$\{5, 3, 3\}$	45

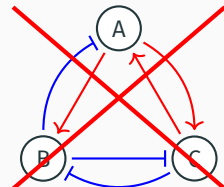
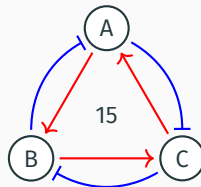


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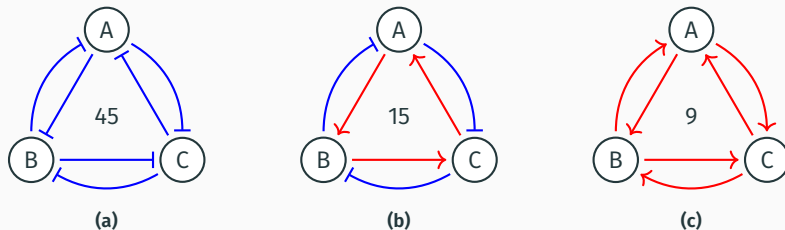


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²Bv et al., *bioRxiv*, 2026

EQUIPOTENT 4-NODE ARE DEGENERATE

m	$q(m)$		Π
0	$\{\kappa_0, \kappa_2, \kappa_2, \kappa_2\}$	$\{1, 14, 14, 14\}$	2744
1	$\{\kappa_1, \kappa_3, \kappa_1, \kappa_1\}$	$\{6, 19, 6, 6\}$	4104
2	$\{\kappa_2, \kappa_2, \kappa_2, \kappa_0\}$	$\{14, 14, 14, 1\}$	2744
3	$\{\kappa_3, \kappa_1, \kappa_1, \kappa_1\}$	$\{19, 6, 6, 6\}$	4104

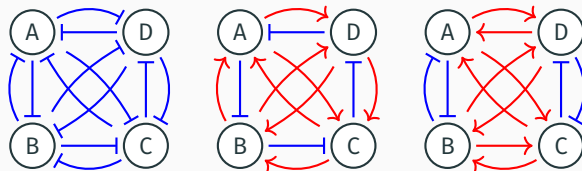


Figure: 4-node equipotent networks with maximum prevalence (4104 MBMs) among equipotent network

¹ m = No. of inhibitions; $q(m)$ = Columns values; Π = row-wise product

²Bv et al., *bioRxiv*, 2026

EQUIPOTENT 4-NODE ARE DEGENERATE

m	$q(m)$		Π
0	$\{\kappa_0, \kappa_2, \kappa_2, \kappa_2\}$	$\{1, 14, 14, 14\}$	2744
1	$\{\kappa_1, \kappa_3, \kappa_1, \kappa_1\}$	$\{6, 19, 6, 6\}$	4104
2	$\{\kappa_2, \kappa_2, \kappa_2, \kappa_0\}$	$\{14, 14, 14, 1\}$	2744
3	$\{\kappa_3, \kappa_1, \kappa_1, \kappa_1\}$	$\{19, 6, 6, 6\}$	4104

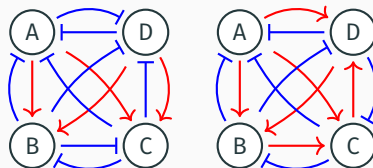


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²Bv et al., *bioRxiv*, 2026

EQUIPOTENT 4-NODE ARE DEGENERATE

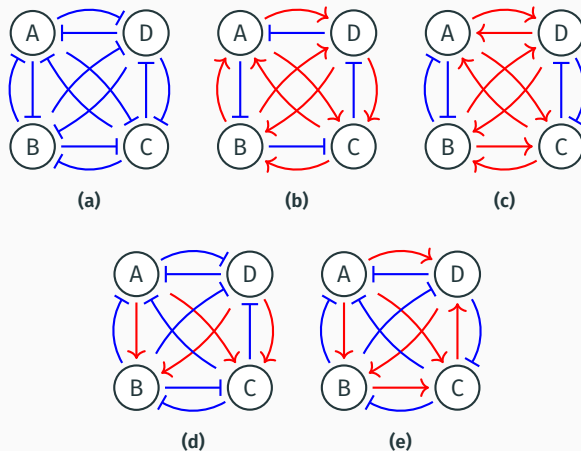


Figure: 4-node equipotent networks with maximum prevalence (4104 MBMs) among equipotent network

¹ m = No. of inhibitions; $q(m)$ = Columns values; $\prod$ = row-wise product

²Bv et al., *bioRxiv*, 2026

NETWORKS WITH 1 INHIBITION EACH HAS THE HIGHEST PREVALENCE AMONG 5-NODE EQUIPOTENT

m	$q(m)$		$\prod$
0	$\{\kappa_0, \kappa_3, \kappa_3, \kappa_3, \kappa_3\}$	$\{1, 148, 148, 148, 148\}$	479, 785, 216
1	$\{\kappa_1, \kappa_4, \kappa_2, \kappa_2, \kappa_2\}$	$\{20, 167, 84, 84, 84\}$	1, 979, 631, 360
2	$\{\kappa_2, \kappa_3, \kappa_3, \kappa_1, \kappa_1\}$	$\{84, 148, 148, 20, 20\}$	735, 974, 400
3	$\{\kappa_3, \kappa_2, \kappa_2, \kappa_2, \kappa_0\}$	$\{148, 84, 84, 84, 1\}$	87, 720, 192
4	$\{\kappa_4, \kappa_1, \kappa_1, \kappa_1, \kappa_1\}$	$\{167, 20, 20, 20, 20\}$	26, 720, 000

¹ m = No. of inhibitions; $q(m)$ = Columns values; $\prod$ = row-wise product

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NETWORKS WITH 1 INHIBITION EACH HAS THE HIGHEST PREVALENCE AMONG 5-NODE EQUIPOTENT



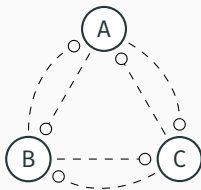
Figure: 5-node equipotent networks with maximum prevalence among equipotent network

¹ m = No. of inhibitions; $q(m)$ = Columns values; $\prod$ = row-wise product

²Bv et al., *bioRxiv*, 2026

WHICH NETWORK(S) OFFER MULTISTABILITY BETWEEN THE SINGLE HIGH STATES?

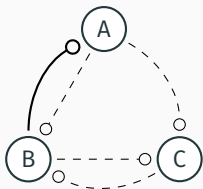
(100), (010), (001)



¹Bv et al., *bioRxiv*, 2026

(100), (010), (001)

$$f_i \in U(\beta_i(0)) \cap L(\beta_i(1)) \cap L(\beta_i(0))$$

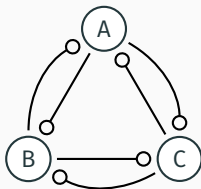


- Non-fully connected:
 f_i gives 0 and 1 for same input

¹Bv et al., *bioRxiv*, 2026

$(100), (010), (001)$

$$f_i \in U(\beta_i(00)) \cap L(\beta_i(10)) \cap L(\beta_i(01))$$

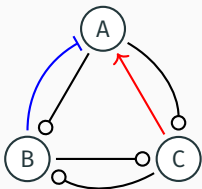


- Non-fully connected:
 f_i gives 0 and 1 for same input
- Network should be fully connected

¹Bv et al., *bioRxiv*, 2026

(100), (010), (001)

$$f_i \in U(10) \cap L(00) \cap L(11) = \emptyset$$

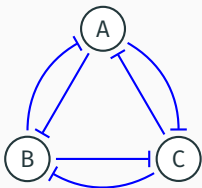


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- Monotonicity requires $b_u \not\leq b_l$

¹Bv et al., *bioRxiv*, 2026

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- Non-fully connected:
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- Network should be fully connected
- Monotonicity requires $b_u \not\leq b_l$
- All the inputs should be inhibitory

¹Bv et al., *bioRxiv*, 2026

$(100), (010), (001)$

- Need not be fully connected

$$f_i \in U(\beta_i(100)) \cap L(\beta_i(010)) \cap L(\beta_i(001))$$

¹Bv et al., *bioRxiv*, 2026

$(100), (010), (001)$

$$f_i \in U(\beta_i(10)) \cap L(\beta_i(01)) \cap L(\beta_i(00))$$

$$\beta_i(e^{i,s}) \not\leq \beta_i(e^{j,s}) \quad \text{and} \quad \beta_i(e^{i,s}) \not\leq \beta_i(\mathbf{0}^s)$$

- Need not be fully connected
- Self-activation: Any input sign β_i satisfies this
- Self-inhibition: All the other inputs should be inhibitory

¹Bv et al., *bioRxiv*, 2026

SELF-ACTIVATION OR MUTUAL INHIBITION REQUIRED

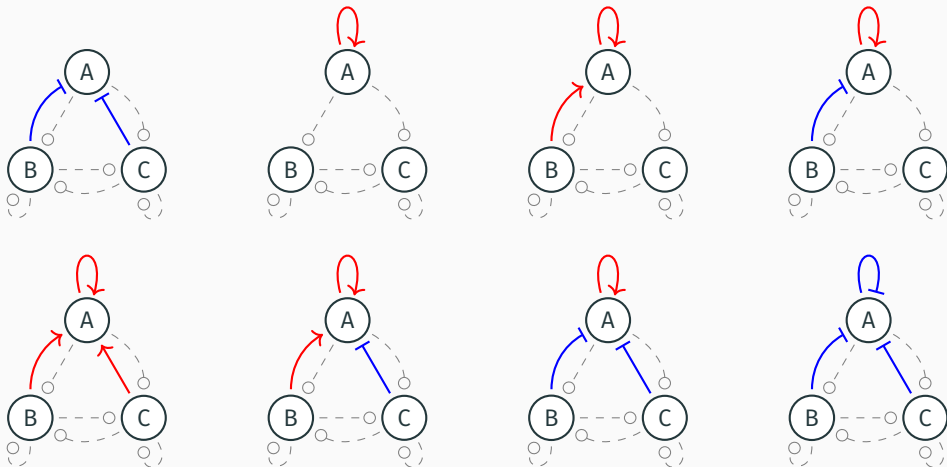


Figure: Input structures compatible with multistability between single high states

**WHICH NETWORK(S) HAS THE HIGHEST PREVALENCE OF MULTISTABILITY BETWEEN THE
SINGLE HIGH STATES?**

COUNTING MULTISTABLE FUNCTIONS

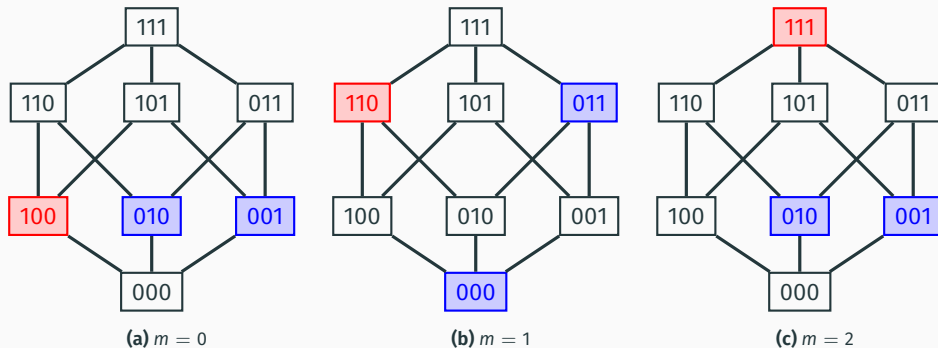


Figure: Counting multistable functions by constraining the Boolean Lattice

¹ m = No. of inhibitions. Blue - $f_i(b) = 0$, Red - $f_i(b) = 1$; Shaded - direct constraints, Outline - from monotonicity.

²Bv et al., *bioRxiv*, 2026

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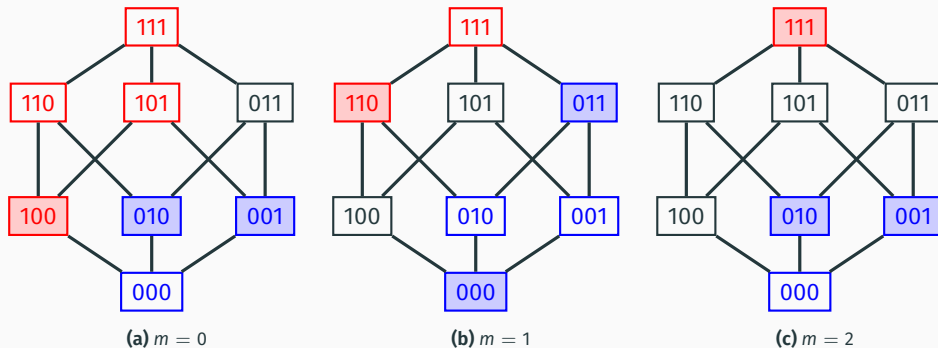


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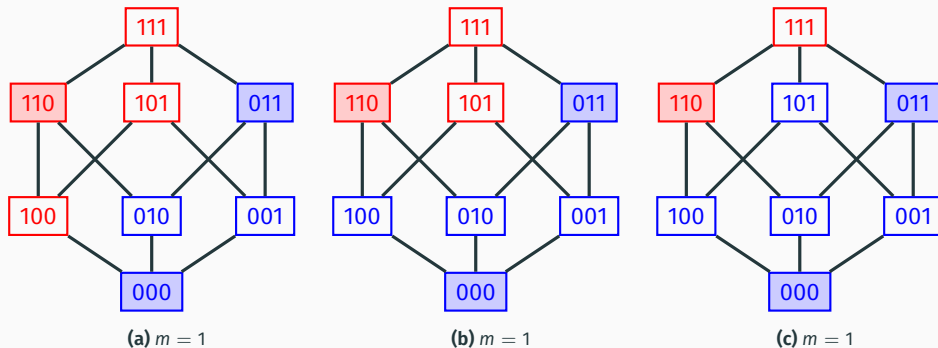


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² Bv et al., *bioRxiv*, 2026

SELF-ACTIVATION AND MUTUAL INHIBITION DOES NOT LEAD TO MOST MULTISTABLE PREVALENCE

n	3			4				5				
m	0	1	2	0	1	2	3	0	1	2	3	4
$ W_i $	2	3	10	9	17	14	38	114	575	445	148	334

Table: Number of MBFs supporting multistability between single high states

¹ n = No. of nodes; m = No. of inhibitions; $|W_i|$ = No. of multistable MBFs

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CONCLUSION

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- Multistable single-high
 - $\iff$ Fully connected mutually inhibitory (or) self-activation
 - Both $\neq$ Maximum prevalence
 - Open problem for maximum prevalence

ACKNOWLEDGEMENTS



THANK YOU

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